Extra Credit Assignment: Airline Market Entry Analysis

Due Date: 12/17/2025

Grading Policy: Successful completion of this project can add up to a 25% boost to your total Midterm Exam score (will not exceed 100). This is a comprehensive challenge estimated to take approximately 8 hours to complete.

Problem Statement: You have been hired as a Data Analyst for an airline company looking to enter the United States domestic market. The company has decided to start with 5 round-trip routes between Medium and Large US airports.

Your task is to analyze Q1 2019 data to identify the best routes for investment. You must demonstrate skills in Data Management (cleaning/merging), Builder Mindset (efficient/reusable code), and Business Intent (actionable insights).

Datasets Provided:

- **Flights**: Contains route availability, distance, delays, and occupancy rates.
- **Tickets**: Contains sample ticket prices (use this for revenue calculations, but do not use it for occupancy).
- **Airport Codes**: Identifies airport size (Medium/Large) and location.

Part 1: Data Preparation & Cleaning

- **Filter**: Consider only round-trip routes between Medium and Large airports.
- **Cancellations**: Exclude canceled flights from all calculations.
- **Metadata**: If you create new fields (columns), you must provide metadata defining them (e.g., in Python docstrings or a separate table).
- **Assumptions**: Real-world data is messy. If you are uncertain of a "right" answer, use your best judgment, make a documented assumption, and proceed.

Part 2: The Analysis

You must answer the following four questions using the logic provided below:

1. **The Busiest Routes:** Identify the 10 busiest round-trip routes based on the number of round-trip flights in the quarter.
2. **The Most Profitable Routes:** Identify the 10 most profitable round-trip routes (ignoring upfront aircraft costs). You must calculate Profit, Total Revenue, and Total Cost.
    a. **Revenue Logic:**
        i. Ticket Revenue: Based on the Tickets dataset.
        ii. Baggage Revenue: $35 per checked bag. Assume 50% of passengers check 1 bag per flight (Total $70 per passenger for a round trip).
        iii. Occupancy: Use the OCCUPANCY_RATE from the Flights dataset. Each plane accommodates up to 200 passengers.
    b. **Cost Logic:**
        i. Flight Ops: Fuel, Oil, Maintenance, Crew = $8.00 per mile.
        ii. Overhead: Depreciation, Insurance, Other = $1.18 per mile.
        iii. Airport Fees: $5,000 for Medium airports; $10,000 for Large airports. (Charged once per landing; a round trip has two landings).
        iv. Delay Costs: The first 15 minutes of departure or arrival delay are free. Afterwards, it costs $75 per minute.
3. **Strategic Recommendation:**
    a. Recommend 5 specific round-trip routes to invest in.
        i. These do not strictly have to be the most profitable; you may base this on any factors you choose, provided you justify them.
        ii. Identify the Origination and Destination airports for each.
    b. **Break-Even Analysis:** The company must acquire 5 new airplanes (one per route) at an upfront cost of $90 million each.
        i. Calculate the number of round-trip flights required to break even on the upfront airplane cost for each of your 5 recommended routes.
        ii. Recommend Key Performance Indicators (KPIs) to track in the future to measure success.

Part 3: Visualization & Reporting

Do not just provide code; you must tell a story.

- **Visuals**: Create charts/plots to support your narrative (e.g., showing key metric drivers). Omit charts that do not add value.
- **Next Steps**: Briefly explain what you would do next if you had more time (e.g., additional data sources or models).

Submission Requirements:

- Submit the following files:
  - Source Code: Working Python code (Jupyter Notebook preferred). Code must be modular, commented, and efficient.
  - Documentation: A summary of your findings, metadata, and assumptions.
  - Visualizations: Embedded in the notebook.